

KAARIGAR



A Portable Professional Identity, Earnings and
Fair-Pricing Platform for India's Skilled Trade Workers

System & Feature Architecture Document

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1. Executive Summary

Kaarigar is an Android-first, offline-capable platform that gives India's skilled blue-collar tradespeople - electricians, plumbers, carpenters, masons, painters, welders, AC technicians - three things they currently do not have: **a portable, verifiable record of their skill and work history**, **a structured record of what they earn**, and **access to transparent market rates** so that pricing stops being a guessing game weighted against them.

The product is built around one central object: the **Digital Kaarigar Passport**. Every other module writes into it. Completing a job adds a verified portfolio entry. Logging payment builds an income record. Receiving a review adds a reputation signal. The passport is owned by the worker, exportable, and readable by anyone through a QR code, which means the worker's credibility is not locked inside our platform - it travels with them.

This is the strategic difference from existing aggregators. Marketplaces such as Urban Company or JustDial own the customer relationship, take a 20-30% commission, and hold the worker's ratings hostage inside their walled garden. Kaarigar takes **zero commission from workers** and treats reputation as the worker's property, not the platform's asset.

ARCHITECTURAL THESIS

Identity is the product; the dashboard, pricing engine, and review system are instruments that generate verifiable evidence for that identity. Design every module to answer: "what provable claim does this add to the worker's passport?"

2. Problem Context

2.1 The four failures we address

Failure	What it looks like on the ground	Kaarigar's response
Invisible skill	A mason with 15 years of experience has no way to prove it to a customer in a new locality. Trust restarts from zero on every move.	Verifiable credentials + geotagged work portfolio + QR-shareable public passport.
Invisible income	Cash-based, unrecorded earnings mean no credit history, so no formal loan for tools, vehicle, or a child's education. Credit comes from informal lenders at punitive rates.	Structured earnings ledger that produces a bank-presentable income statement, shared only with explicit consent.
Opaque pricing	The worker does not know the going rate; the customer does not either. Negotiation is adversarial and information-poor. Middlemen exploit the gap on both sides.	Fair Price Engine publishing a data-backed rate band per task, city, and complexity, plus an itemised quote builder.
Trapped reputation	Word-of-mouth does not scale beyond a few streets and cannot be carried to a new city or platform.	Job-anchored, portable reputation issued as a signed credential.

2.2 Policy alignment (useful for the SIH evaluation)

- **PM Vishwakarma Yojana** - directly targets traditional artisans and craftspeople; Kaarigar is a digital delivery layer for exactly this cohort.
- **e-Shram** - the national unorganised-worker database; passport can carry the e-Shram UAN as a verified attribute.
- **DigiLocker / NSDC / Skill India Digital** - source of authentic ITI, PMKVY and NSQF certificates.
- **ONDC** - a future path for service discovery without platform lock-in.
- **DPDP Act 2023** - consent, purpose limitation and data minimisation are designed in, not retrofitted.

3. Users, Roles and Permissions

Role	Primary needs	Access scope
Kaarigar (worker - primary user)	Log earnings, get fair rates, prove skill, find work, track dues.	Full read/write on own passport, ledger, jobs; read-only on rate bands; no access to other workers' data.
Customer	Find a trustworthy nearby worker, get a clear quote, pay, review.	Read public passports; write reviews only for jobs they booked; read own job history.
Contractor / Team Lead	Assemble crews, assign jobs, split payouts.	Read crew members' availability and job status with consent; write crew assignments and endorsements.
Verifier (ITI, NGO, skill council, senior tradesperson)	Attest a worker's skill or certificate.	Write attestations only; can revoke own attestations; no ledger access, ever.
Institution / Admin (district admin, NGO, skill mission)	Aggregate view of skill supply, income levels, training gaps.	Aggregated and anonymised analytics only. No individual-level PII, no individual earnings.
Platform Ops	Dispute resolution, fraud review, content moderation.	Scoped, fully audit-logged access under a documented break-glass procedure.

4. Feature Architecture

Eight functional modules. Each is described by its purpose, mechanics, and the evidence it contributes to the passport. Tags mark scope: **MVP** for the hackathon build, **V1** post-hackathon, **Later** future roadmap.

Module A - Digital Kaarigar Passport **MVP**

Purpose. A single, portable, verifiable professional identity. This is the anchor object of the entire system.

Composition

- **Base identity** - mobile number (OTP-verified) as the primary key; optional Aadhaar verification through DigiLocker or offline eKYC XML. We store a verification token and masked last-four digits only; **the raw Aadhaar number is never persisted**.
- **Trade taxonomy** - primary trade plus sub-skills, coded against NCO-2015 / NSQF so the data is interoperable with government systems rather than free text.
- **Credentials** - ITI, PMKVY, NSDC certificates fetched from DigiLocker; each stored with issuer, issue date, and verification status.
- **Work portfolio** - before/after photographs from completed jobs, captured in-app with geotag and timestamp so authenticity is verifiable, not merely claimed.
- **Endorsements** - structured attestations from customers, contractors, and verifiers.
- **Public passport page** - a server-rendered page at `kaarigar.in/k/{handle}` with a printable QR code, so a customer can verify a worker without installing anything.

Verifiable credential design

On any significant event (certificate verified, 25th job completed, endorsement received), the platform issues a W3C Verifiable Credential signed with the platform key, stored on the worker's device and mirrored server-side. A third party can validate the signature offline. This is what makes the reputation genuinely portable rather than a screenshot.

Module B - Kamai: Earnings & Ledger **MVP**

Purpose. Convert invisible cash income into a structured, defensible financial record.

Mechanics

- **One-tap job entry** - trade icon, customer, amount received, advance vs balance, material cost. Target: under 15 seconds, usable one-handed, at a worksite, in sunlight.
- **Udhaar tracker** - outstanding receivables per customer, with a polite reminder the worker can dispatch over WhatsApp or SMS in one tap. This solves a problem workers cite constantly: money owed and forgotten.
- **Material expense capture** - photograph a shop bill; OCR extracts amount and date; the worker confirms. Enables true net-margin calculation instead of gross-receipts illusion.
- **Summaries** - day, week, month and season views rendered as icon-and-number cards with large numerals, plus a comparison against the worker's own past periods (never against other workers, which would be demoralising and privacy-invasive).
- **Income statement export** - a clean PDF of 6- or 12-month income suitable for a loan or subsidy application.

Ledger integrity

The ledger is **append-only**. Corrections are recorded as reversing entries with a reason code rather than in-place edits. This preserves an audit trail, makes the export credible to a lender, and eliminates most offline-sync conflicts by construction.

Module C - Mol-Bhav: Fair Price Engine **MVP**

Purpose. Replace adversarial guesswork with a shared, data-backed reference price. It protects the worker from undercharging and the customer from being overcharged - which is why it is defensible to both sides.

Mechanics

- **Rate band, not a rate.** For each (trade, task, city tier, locality, complexity) the engine publishes a 25th-75th percentile band, plus the median. A single number would be false precision and would invite price-fixing criticism; a band is honest about variance.
- **Cold-start seeding.** Before we have transaction volume, bands are seeded from published government schedules of rates (CPWD DSR and state PWD schedules) and state minimum-wage notifications for skilled labour. This gives day-one credibility.
- **Live correction.** As jobs complete, actual settled prices feed back as observations, re-fitting the band per locality. Observations are weighted by verification strength so an unverified self-report cannot skew a band.
- **Quote builder.** The worker picks line items; the app assembles an itemised estimate (labour, material, visit charge, timeline) and produces a shareable link or PDF. An itemised quote is itself a professionalism signal that raises the worker's perceived standing.
- **Negotiation view.** Shows the quoted price against the band with plain-language justification points the worker can use ("this task needs conduit chasing, which is why it sits above the median").

DESIGN GUARD

The engine must never display a recommendation that pushes a worker below the applicable statutory minimum wage. Enforce this as a hard floor in `pricing-svc`, not as a UI suggestion.

Module D - Reputation & Discovery MVP

Purpose. Make local trustworthiness measurable, portable and hard to fake.

Mechanics

- **Job-anchored reviews only.** A review can be written only against a job record that exists, was accepted, and was marked complete. This single constraint eliminates the large majority of fake-review attacks that plague open review systems.
- **Multi-dimensional rating** - workmanship, punctuality, price honesty, cleanliness. A composite hides useful information; four axes let a customer weigh what matters to them.
- **Hyperlocal ranking** - discovery is weighted by distance (3-8 km default radius), availability today, rate band fit, and rating recency. A worker four kilometres away who can come today usually beats a marginally better-rated worker across the city.
- **Review lifecycle** - workers may post a public response; disputed reviews go to a review queue rather than silent deletion.

Module E - Job & Work Management MVP

Purpose. Give the job a lifecycle, because the lifecycle is what generates verified evidence.

State machine: REQUESTED → QUOTED → ACCEPTED → SCHEDULED → IN_PROGRESS → COMPLETED → SETTLED → REVIEWED, with CANCELLED and DISPUTED as terminal branches. Every transition is timestamped and immutable.

- **Availability toggle** - a single "available today" switch driving discovery ranking. Complex calendars fail with this user group; a binary switch does not.
- **Job card** - scope, agreed price, before/after photographs, materials used. Completion auto-populates the portfolio and prompts the ledger entry.
- **Crew mode V1** - a contractor assigns helpers to a job and records the payout split, so helpers also accumulate verified job history instead of being invisible.

Module F - Financial Inclusion Bridge V1

Purpose. Turn the income record into access to formal finance. **Kaarigar is not a lender and does not underwrite.** It is a consent-mediated data bridge.

- **Credit-readiness view** - shows the worker what a lender would see: income regularity, growth trend, job volume, verification depth. Framed as "here is what strengthens your case", never as a rejection.
- **Consent artifact** - modelled on the Account Aggregator / DEPA pattern. Every share carries an explicit purpose, data scope, expiry date, and one-tap revocation, and is written to an immutable consent log.
- **Scheme discovery** - eligibility nudges for PM Vishwakarma, PM-SYM pension, e-Shram registration, and tool-purchase subsidies.
- **Micro-insurance and tool EMI Later** - partner-delivered, never platform-underwritten.

Module G - Skilling & Growth Later

- Short vernacular video modules per trade, weighted toward safety (electrical safety, working at height, tool handling) - the highest-value content for this cohort.
- Gap-based suggestions: if a worker declines or never receives a common task type in their locality, suggest the relevant module.
- Completion writes a credential into the passport and links out to formal NSDC/PMKVY certification pathways.

Module H - Access Layer (cross-cutting) MVP

This module is not a feature list so much as a constraint set that governs every screen in the product.

- **Icon-first, low-literacy UI** - every action carries an icon and a number; text is supporting, not load-bearing. Tap targets at least 48dp. High contrast for outdoor sunlight.
- **Multilingual** - Hindi and English at MVP; Punjabi, Marathi, Bengali, Tamil, Telugu, Kannada, Gujarati, Odia, Malayalam, Assamese in V1.
- **Offline-first** - full read and write capability with no network; see Section 9.
- **Low-end device budget** - APK under 25 MB, functional on Android 8 with 2 GB RAM.
- **WhatsApp channel** - job alerts, quote sharing, and payment reminders through the WhatsApp Business Cloud API, which is where this user group already lives.
- **SMS/IVR fallback** Later - for feature-phone users, covering job alerts and balance queries.
- **Text-to-speech readout** Later - optional accessibility aid for numbers and summaries. Deliberately scoped as an accessibility helper only; the earlier voice-first framing was dropped and should stay out of the MVP, as it multiplies build risk for a demo.

5. System Architecture

5.1 Layered view



5.2 Why a modular monolith, not microservices

The service boundaries above are real and enforced at the module level, but they deploy as a single FastAPI application backed by one MongoDB replica set. For a hackathon and an early pilot this is the correct choice: one deployment target, no distributed-transaction

problems, no inter-service network latency, and a team of four can actually run it. Because the boundaries are drawn now - separate routers, separate data access modules, no cross-module direct database reads - any service can be extracted into its own process later without rewriting business logic. Choosing microservices at this stage would trade real delivery speed for imaginary scale.

6. Technology Stack

Layer	Choice	Rationale
Mobile	React Native (Expo) + TypeScript	One codebase for worker and customer apps; JS/TS reuse with the Next.js web surfaces; large offline-storage ecosystem. Flutter is a defensible alternative if the team is stronger in Dart - decide on team skill, not benchmarks.
On-device DB	WatermelonDB (SQLite)	Built for offline-first sync with an observable query layer; handles thousands of records on low-end hardware without UI jank.
Web	Next.js (App Router)	SSR for the public passport page - matters for shareability and search visibility; same framework for the admin console.
Backend	FastAPI (Python 3.11)	Async I/O, automatic OpenAPI documentation, Pydantic validation at the boundary, and native proximity to the ML components.
Database	MongoDB 7 replica set	Flexible documents suit heterogeneous trade profiles and portfolio items; 2dsphere geo index serves discovery directly; multi-document transactions cover ledger writes.
Cache / Queue	Redis 7 + Celery	OTP store, hot rate bands, rate limiting, and the async worker backbone.
Object storage	MinIO (dev) / S3 (prod)	Media and generated PDFs kept out of the database; presigned URLs for direct upload from the device.
ML	scikit-learn / LightGBM, ONNX Runtime	Gradient-boosted trees fit tabular pricing data better than deep models at this data scale and stay explainable. ONNX for any on-device inference.
OCR	PaddleOCR or Tesseract + Indic models	Bill extraction; runs server-side in a worker to keep the APK small.
Auth	Phone OTP + JWT (access/refresh), device binding	Passwords are a barrier for this user group. OTP is the norm they already understand.
Infra	Docker Compose -> single VPS; K8s-ready manifests	Demo runs on one machine; the same images lift to a cluster unchanged when volume justifies it.
CI/CD	GitHub Actions	Lint, test, build, push image, deploy on merge to main.
Observability	Prometheus + Grafana, Sentry	API latency and error budgets; crash reporting from field devices on poor networks.

7. Data Model

Principal collections and their defining fields. Identifiers are client-generated UUIDv7 so that records created offline keep a stable, sortable identity from the moment of creation.

Collection	Key fields	Indexes / notes
users	_id, phone, role[], languages, created_at, device_ids[], status	Unique index on phone.
kaarigar_profiles	user_id, display_name, trades[{nco_code, years_exp, level}], service_area{geo, radius_km}, availability_today, trust_score{value, components{}}, computed_at, passport_handle	2dsphere on service_area.geo; unique on passport_handle.
credentials	profile_id, type, issuer, issued_on, source (digilocker/manual/attestation), verification_status, evidence_url, vc_jwt	Compound (profile_id, verification_status).
portfolio_items	profile_id, job_id, media[{url, phash, geo, captured_at}], trade, description, visibility	phash indexed for duplicate-media detection.
jobs	_id, customer_id, kaarigar_id, trade, tasks[], state, state_history[{state, at, by}], agreed_price, location{geo, locality}, crew[]	Compound (kaarigar_id, state); 2dsphere on location.geo.
quotes	job_id, line_items[{task_code, qty, unit_rate, source}], labour_total, material_total, valid_until, band_snapshot	band_snapshot preserves what the engine advised at quote time.
ledger_entries	_id (client UUID), profile_id, job_id?, direction (in/out), amount, category, payment_mode, occurred_on, reverses_id?, sync_state	Append-only. Compound (profile_id, occurred_on). Idempotent on _id.
receivables	profile_id, customer_ref, job_id, amount_due, due_on, reminders[], settled_at	Derived view maintained from ledger events.
reviews	job_id (unique), author_id, subject_id, ratings{workmanship, punctuality, price_honesty, cleanliness}, text, response, flags[]	Unique on job_id enforces one review per job.
rate_bands	trade, task_code, city_tier, locality, p25, p50, p75, sample_n, wage_floor, seeded_from, updated_at	Compound (trade, task_code, locality). Cached in Redis.
price_observations	job_id, task_code, settled_price, locality, verification_weight, observed_at	Feeds nightly band re-fitting.
consents	subject_id, grantee, purpose, data_scope[], granted_at, expires_at, revoked_at, artifact_hash	Immutable once written; revocation is a new event.
disputes	job_id, raised_by, category, evidence[], state, resolution, resolved_by	Freezes affected review and score contribution while open.
audit_log	actor_id, action, target, ip, at, meta	Write-once; retained separately from operational data.

8. API Surface (representative)

Method	Endpoint	Purpose
POST	/auth/otp/request	Issue OTP to a phone number (rate-limited).
POST	/auth/otp/verify	Verify OTP, bind device, return access + refresh tokens.
GET / PATCH	/passport/me	Read and update the authenticated worker's passport.
POST	/passport/credentials/digilocker	Initiate DigiLocker fetch and import certificates.
GET	/p/{handle}	Public passport view (no auth; only fields the worker marked public).
POST	/ledger/entries	Create a ledger entry. Idempotent on client UUID.
POST	/ledger/entries/{id}/reverse	Post a reversing entry with a reason code (no in-place edits).
GET	/ledger/summary?period=month	Aggregated income, expense, net, and receivables.
POST	/ledger/export/income-statement	Queue generation of a lender-presentable income PDF.
GET	/pricing/band?trade&task&lat&lng	Return p25/p50/p75, sample size, and the statutory wage floor.
POST	/quotes	Build an itemised quote; snapshots the band used.
POST	/jobs /jobs/{id}/transition	Create a job; move it through the lifecycle state machine.
GET	/discovery/search?trade&lat&lng&radius	Ranked nearby workers with availability and band fit.
POST	/reviews	Submit a review; rejected unless the job is COMPLETED and the author is its customer.
POST	/sync/batch	Offline outbox flush: ordered, idempotent, returns per-item results.
POST / DELETE	/consents	Grant or revoke a scoped, time-bound data share.

9. Offline-First and Sync Design

This is the single most important engineering decision in the product. The user works in basements, stairwells, under-construction sites, and villages with intermittent 2G. An app that requires connectivity to log a payment will be abandoned within a week.

9.1 Principles

- **The device is the source of truth for creation.** Every write succeeds locally first and enters an outbox. The UI never blocks on the network and never shows a spinner for a local action.
- **Client-generated UUIDv7 identifiers.** Records have a permanent, sortable ID from creation, so nothing needs renumbering after sync.
- **Idempotent ingest.** /sync/batch is safe to replay; the server deduplicates on client UUID. A flaky connection that half-uploads a batch causes no duplicates.
- **Append-only ledger removes most conflicts.** Two devices cannot disagree about an immutable entry. Corrections are new entries, so the merge is a union, not a resolution.
- **Conflict policy for mutable documents** (profile fields, availability): last-write-wins by server receipt time, with the losing version retained in a shadow field so nothing is silently destroyed.
- **Media deferred.** Photographs queue separately and upload only on unmetered or strong connections; the job record syncs immediately without waiting for its images.
- **Visible sync state.** Every record shows pending / synced / failed with a simple icon. Hidden sync erodes trust in a financial record; workers must be able to see that their money data is safe.

9.2 Sync sequence

1. Local write commits to SQLite and appends to the outbox.
2. Connectivity listener or a periodic job wakes the sync engine.
3. Outbox flushes in causal order (jobs before their ledger entries before their reviews) in batches of up to 50.
4. Server validates, applies idempotently, and returns per-item accept/reject with reasons.
5. Accepted items are marked synced; rejected items surface a specific, actionable message rather than a generic failure.
6. A delta pull follows, using a server watermark, to bring down rate-band updates, new job requests, and reviews.

10. AI / ML Components

Component	Approach	Notes and honest limitations
Fair price model	LightGBM regression on trade, task code, city tier, locality, area income proxy, materials, urgency, worker experience, season; quantile regression for p25/p75.	Useless without volume. At MVP this is a statistical band over seeded government rates and any collected observations - present it honestly as such. Fabricating an "AI pricing model" over 40 data points is the kind of claim a sharp judge will puncture.
Fake-review detection	Graph and behavioural anomaly detection: device reuse, review bursts, reviewer-worker geo mismatch, account age, rating-pattern uniformity.	Flags for human review; does not auto-delete. Job-anchoring already blocks most naive attacks, so this is a second line.
Portfolio media integrity	Perceptual hashing to detect the same photograph claimed by multiple workers; EXIF and geotag consistency against the job location.	Catches image theft from the internet or from other workers. Cheap, deterministic, effective - no model training required.
Bill OCR	PaddleOCR with Indic script support; regex and layout heuristics for total, date, vendor.	Always confirmed by the user before the entry is committed. Never silently write a financial record from an OCR guess.
Trust score	Transparent weighted rubric - deliberately not a learned model. See Section 11.	A black-box model governing someone's livelihood is indefensible. Explainability is a requirement, not a nice-to-have.
Demand forecasting Later	Seasonal decomposition on locality job volume by trade.	Tells a worker when demand for their trade rises so they can plan. Needs a year of data before it means anything.

11. Trust Score Design

The score is a published rubric, computed nightly, with every component visible to the worker alongside a specific action to improve it.

Component	Weight	Basis
Identity verification	20	Phone verified, eKYC completed, e-Shram UAN linked.
Skill credentials	15	Verified certificates and verifier attestations, weighted by issuer authority.
Verified work history	25	Count and recency of COMPLETED jobs with photographic evidence; logarithmic so volume cannot dominate.
Customer ratings	25	Bayesian-adjusted mean across the four rating axes, so three glowing reviews do not outrank forty solid ones.
Reliability record	-10 to 0	Late cancellations, no-shows, and upheld disputes.
Skilling engagement	5	Completed safety and upskilling modules.

Fairness safeguards

- **New-worker floor.** A new joiner starts at a neutral baseline, not at zero. Starting at zero would make the platform useless to precisely the people who need it most.
- **Rate-limited decline.** The score cannot fall by more than a fixed amount per week, so one bad customer cannot destroy a livelihood overnight.
- **Appeal path.** Every negative component is contestable through the dispute flow, and the contested component is frozen while under review.
- **No protected proxies.** Caste, religion, region of origin, and gender are never collected and never enter the score. Locality is used for discovery distance only, never as a quality signal.

12. Security, Privacy and Compliance

12.1 Controls

- **Authentication** - phone OTP with rate limiting and device binding; JWT access tokens (15 min) plus rotating refresh tokens; refresh reuse detection revokes the family.
- **Authorisation** - role-based access enforced at the service layer, never only in the UI. Object-level checks on every ledger and passport read.
- **Aadhaar handling** - offline eKYC XML or DigiLocker only. Store a verification token, name-match result, and masked last-four. **The raw number is never written to disk or logs.**
- **Encryption** - TLS 1.3 in transit; encryption at rest on database volumes; field-level encryption for contact details and eKYC artifacts.
- **Media** - presigned, short-lived upload and download URLs; private buckets; portfolio visibility controlled per item by the worker.
- **Audit** - append-only log for every access to another user's data, every consent event, and every ops break-glass action.
- **Secrets** - environment-injected, never committed; key rotation documented for the VC signing key in particular.

12.2 DPDP Act 2023 alignment

Principle	Implementation
Purpose limitation	Every consent artifact names a specific purpose; the ledger cannot be read for any purpose not enumerated in a live consent.
Data minimisation	No caste, religion, or income-source profiling. Location captured at locality granularity for discovery, precise coordinates only on an active job.
Consent and revocation	Granular, time-bound, one-tap revocable; revocation propagates to partners within a defined window.
Data portability	Full passport and ledger export as signed JSON plus PDF, on demand, without support intervention.
Erasure	Account deletion removes PII and media; anonymised price observations may be retained, which is disclosed up front.
Notice	Pictorial, vernacular consent screens. A dense English privacy policy is not meaningful notice for this user group and should not be treated as compliance.

13. Non-Functional Requirements

Attribute	Target	Reasoning
Offline endurance	7 days of full functionality	Covers a migrant worker on a site with no usable signal for a week.
Sync on 2G	Typical day's outbox under 30 s	Batch compression; media deferred to better connections.
API latency	p95 under 300 ms (excluding media)	Discovery and pricing must feel instant during a live negotiation.
APK size	Under 25 MB	Storage pressure and data cost on entry-level devices.
Device floor	Android 8, 2 GB RAM	Reflects the actual device base of the target user.
Cold start	Under 2.5 s on the device floor	Slow launch is the most common reason low-end users abandon an app.
Availability	99.5% for the API	Offline-first absorbs short outages, which is what makes this target realistic.
Accessibility	48dp targets, 4.5:1 contrast, icon-first	Low literacy, outdoor use, older users, work-worn hands.
Localisation	2 languages MVP, 12 in V1	Language is the binding constraint on adoption outside metros.

14. Build Roadmap

14.1 Hackathon build (what to actually ship in the finals)

The temptation is to build all eight modules shallowly. Resist it. A demo that shows one complete, believable loop beats eight half-features every time.

THE DEMO LOOP TO BUILD END TO END

Worker onboards and imports one certificate → customer searches nearby and finds the worker → worker builds an itemised quote against the fair-price band → job is accepted, completed with before/after photographs → payment logged in the ledger → customer leaves a job-anchored review → passport and trust score update live → the QR passport is scanned on a second phone → a 6-month income statement PDF is generated. **Then pull the network cable, repeat the ledger entry offline, reconnect, and show it sync.** That last step is the one judges will remember.

Priority	Scope	Status for the demo
Must	Auth, passport, jobs lifecycle, ledger, rate band lookup, quote builder, reviews, discovery, offline sync	Fully built and working.
Should	QR public passport page, income statement PDF, trust score with visible breakdown	Built; these are high-impact on stage.
Simulate	DigiLocker fetch, WhatsApp dispatch, UPI settlement	Sandbox or stubbed with a clearly labelled mock. State plainly that it is stubbed - claiming a live government integration you do not have is the fastest way to lose a panel's trust.
Exclude	Skilling module, crew mode, IVR, insurance, lending	Roadmap slide only.

14.2 Beyond the hackathon

Phase	Horizon	Focus
Pilot	Months 1-3	One city, two or three trades, 200-500 workers recruited through an NGO or trade union partner. Objective is retention and data density, not headline user count.
V1	Months 4-8	Full language set, DigiLocker and e-Shram integrations live, crew mode, WhatsApp channel, first lender consent pilot.
Scale	Months 9-18	Multi-city rate bands, skilling content, institution dashboards, ONDC discovery exploration, service extraction from the monolith where load demands it.

15. Differentiation and Sustainability

15.1 Against existing players

Dimension	Urban Company / JustDial model	Kaarigar
Commission	20-30% of job value taken from the worker	Zero commission from workers
Reputation ownership	Held by the platform; lost on exit	Held by the worker as a signed, exportable credential
Customer relationship	Owned by the platform; direct contact discouraged	Owned by the worker; the platform is infrastructure
Pricing	Set by the platform	Negotiated by the parties, informed by a transparent public band
Financial identity	Not a goal	Central - the income record is the point
Offline capability	Effectively none	First-class requirement

15.2 Revenue without taxing the worker

- Institution and government dashboards (anonymised skill-supply and income analytics) on subscription.
- Referral fees from NBFCs and insurers on consented, worker-initiated introductions - paid by the institution, never by the worker.
- Optional premium features for contractors managing crews at scale.
- CSR and skill-mission grants during the pilot phase.

BE READY FOR THIS QUESTION

A judge will ask how you get workers to log cash income they may prefer to keep unrecorded. The honest answer is that the ledger has to pay for itself immediately through the udhaar tracker - workers adopt it because it tells them who owes them money, and the credit history is a by-product they come to value later. Leading with "build your credit score" will not drive adoption; leading with "never lose track of money owed to you" will.

16. Risks and Mitigations

Risk	Severity	Mitigation
Cold start - no workers means no customers, and vice versa	High	Seed supply first through NGO, ITI and union partnerships in one locality; the passport and ledger have standalone value even with zero customers on the platform, which is why they are the wedge.
Rate bands are thin or wrong early on	High	Seed from published government schedules; display sample size honestly; suppress the band entirely below a minimum observation count rather than showing a misleading number.
Fake reviews and reputation gaming	Medium	Job-anchoring, device and graph anomaly detection, rate-limited score movement, human review queue.
Low digital literacy blocks onboarding	High	Icon-first design, assisted onboarding camps with partners, WhatsApp entry path, a design rule that no core task exceeds three taps.
Aadhaar and PII exposure	Critical	Offline eKYC only, no raw storage, field-level encryption, audit logging, external review before any pilot with real users.
Score becomes a gatekeeper that entrenches disadvantage	High	Neutral baseline for new workers, published rubric, appeals, no protected-attribute proxies, periodic disparity audit across localities.
Retention collapses after onboarding drives	Medium	Instrument the udhaar tracker as the retention hook; measure weekly active logging, not registrations.

17. Open Decisions

These need a team call before build starts. They are listed rather than silently assumed.

1. **React Native or Flutter** - decide on the team's existing depth, not framework benchmarks. Either works.
2. **MongoDB for the ledger, or a separate PostgreSQL instance** - Mongo with transactions is sufficient and keeps the stack single. Postgres is the stronger answer if a judge presses on financial data integrity. Choose one and be able to defend it.
3. **Customer app or customer PWA** - a PWA removes an install barrier for a one-off customer and is probably right, but it complicates push notifications.
4. **Pilot city and trades** - narrower is better. Two trades in one city will produce meaningful rate bands; ten trades across five cities will produce noise.
5. **Naming** - "Kaarigar" reads as artisan and aligns with PM Vishwakarma framing; confirm the spelling used in the internal-round submission and keep it consistent across the deck, app, and domain.